



SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Design Finite Automata DFA, NFA, ϵ -NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:50

Design Task

Code of Conduct:

I Peram Vivekananda Reddy(192372330)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Vivek

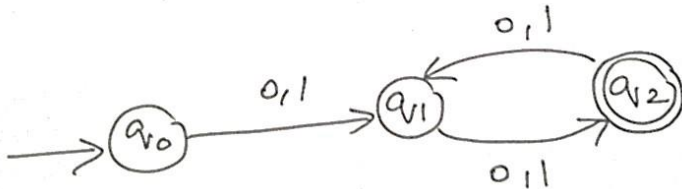


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1. Design the regular expression for the language represented by the given automata



2. Consider the following grammar

$S \rightarrow aSc | T | U | \epsilon$

$T \rightarrow bTc | \epsilon$

$U \rightarrow aUb | \epsilon$

A) What is the language defined by the grammar?

B) Give the derivation tree for the word aaabbc

C) A Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

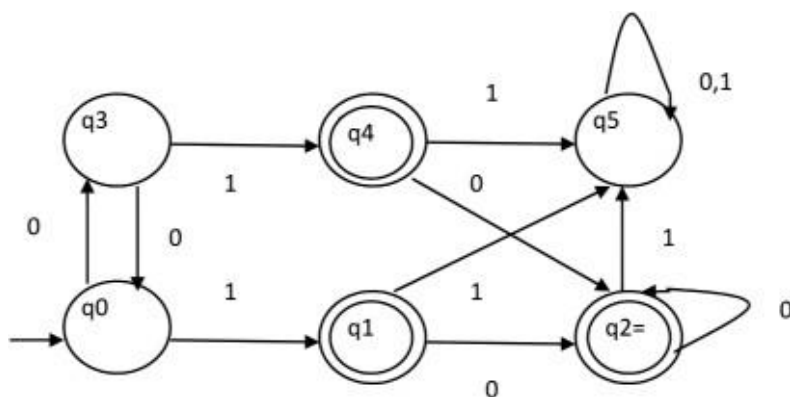
3. Design a non deterministic finite automata with ϵ -transitions to represent the language denoted by the regular expression

$00^*1 + 11^*0$

4. I) Show that if L_1 and L_2 are regular then $(L_1 + L_2)$ also regular

ii) Is the language $\{a^p \mid p \text{ is prime}\}$ regular? Justify

5. Design minimized automata, Consider the DFA given below. Find the states that are equivalent and construct a minimum state automata equivalent to the given automata. Draw the minimized automata, write its transition table and mark initial and final states





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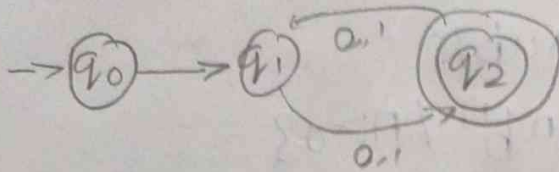


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RUBRICS FOR CALCULATION

Criteria	Weight age	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30	3	3	3	3	3	15
Accuracy of Answer	25	2.5	2.5	2.5	2.5	2.5	12.5
Application of Knowledge	20	2	2	2	2	2	10
Completeness of Response	15	1.5	1.5	1.5	1.5	1.5	7.5
Clarity & Presentation	10	1	1	1	1	1	5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/ 50

① Regular Expression for the Given Automation



From the automation

→ q_0 is the initial state

→ from q_0 , on '0' or '1' $q_0 \rightarrow q_1$

→ from q_1 on '0' or '1', return to q_1 .

→ q_2 is the final state

let : $\Sigma = \{0+1\}$

the regular expression is

$$= (\{0+1\} \{0+1\} \cdot \{0+1\} \{0+1\})$$

$$= (\{0+1\}^2 \{0+1\}^2)$$

Example accepted:

~~0, 01, 011, 10, 11, 00001010~~

00, 01, 10, 11, 0000, 0101, 1010, ...

Given Grammar

$G \rightarrow asc / t / ule$

$T \rightarrow bTC / e \rightarrow auble$

language defined by the grammar

from $t: \gamma + b + c | E$

So, $L(T) = \{b^n c^n \mid n \rightarrow 0\}$

From $U: U + aub/E$

So, $LCU = \{a^n b^n \mid n \geq 0\}$

from s:s $\rightarrow$ ase add same no. of ds at the beginning & is at the end.

$$\therefore L = \{a^i b^j c^k \mid i, j \geq 0 \} \cup \{a^{i+j} b^i c^i \mid i \geq 0\}$$

⑧ deviation free for aaabc

use stage once $S = 195e$

Then : $S \not\models a \cup b$

So $asc \Rightarrow aacabc$

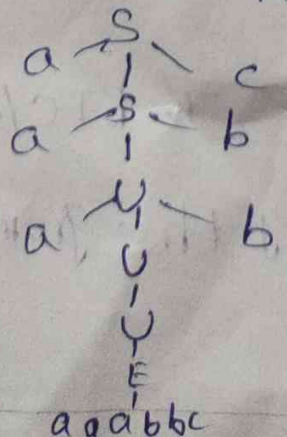
again : $a_u > a_{ub}$

Therefore $\therefore aacubx \rightarrow aacubbc$

finally : $u \rightarrow e$

$$S \Rightarrow 1asc \Rightarrow aac \ bc \Rightarrow aacacbc \Rightarrow aacacbbcc$$

Derivation tree



Step ①: Initial partition

$P_0 = \{ \{q_1, q_2, q_4\}, \{q_0, q_3, q_5\} \}$

Step ②: check equivalent states

q_1, q_2, q_4

for q_1 : $0 \rightarrow q_2$ (final)
 $1 \rightarrow q_5$ (non-final)

for q_2 : $0 \rightarrow q_2$ (final)
 $1 \rightarrow q_5$ (non-final)

for q_4 : $0 \rightarrow q_2$ (final)
 $1 \rightarrow q_5$ (non-final)

$\therefore q_1 = q_2 = q_4$

q_0, q_3

for q_0 : $0 \rightarrow q_3$ for q_3 : $0 \rightarrow q_0$
 $1 \rightarrow q_1$ $1 \rightarrow q_4$

so, $q_0 = q_3$

q_5 remains separate

~~so, q_5~~
 q_5 remains separate

$\therefore$ minimized has 3 states

$\{q_0, q_3\}, \{q_1, q_2, q_4\}, \{q_5\}$

$A = \{q_0, q_3\} \rightarrow$ Initial

$B = \{q_1, q_2, q_4\} \rightarrow$ final

$C = \{q_5\} \rightarrow$ non-final

Transition Table

minimized

$A = \{q_0, q_3\}$

$B = \{q_1, q_2, q_4\}$

$C = \{q_5\}$

A

B

C

0

1

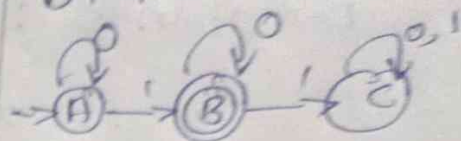
C

C

Initial state = A

Final state = B

DF A (minimized)



$\{ \{q_0, q_3\}, \{q_1, q_2, q_4\}, \{q_5\} \}$

② yes the Grammar is ambiguous

Derivation 1: $S \rightarrow asc = ac$ (inner & belong)

Derivation 2: $S \rightarrow asc$ (inner S can become $\{+ac\}$)

$\therefore \Rightarrow ac$

So, there is two different derivations

$S \rightarrow asc \rightarrow ac$

$S \rightarrow asc \rightarrow asc \rightarrow ac$

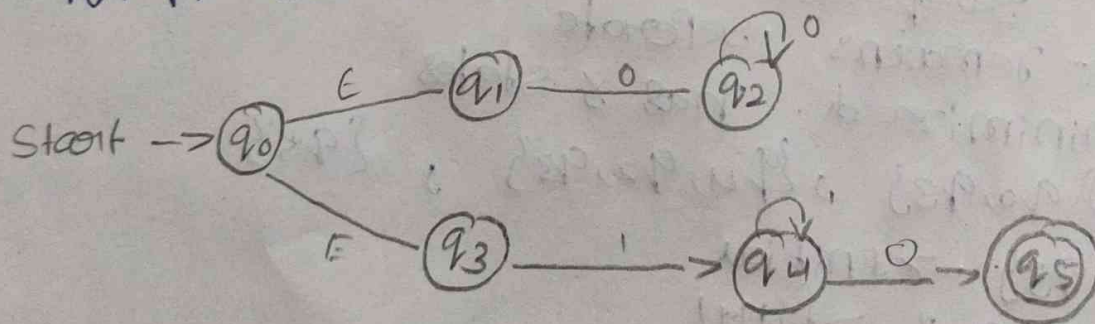
Hence, the grammar is ambiguous

③ NFA with ϵ transitions

Given $RE = 00^* + 11^*0$

Two alternatives are 00^* & 11^*0

NFA Construction



Transition table

84

State	0	1	ϵ
q_0	-	-	q_1, q_3
q_1	q_2	-	-
q_2	q_2	-	-
q_3	-	q_4	-
q_4	-	q_4	-
q_5	-	-	-

1st part: $q_1 \xrightarrow{0} q_2$, $q_2 \xrightarrow{0} q_2$ Sival 00^*

2nd part: $q_3 \xrightarrow{1} q_4$, $q_4 \xrightarrow{1} q_4$, $q_4 \xrightarrow{\epsilon} q_5$

Hence $00^* + 110$.

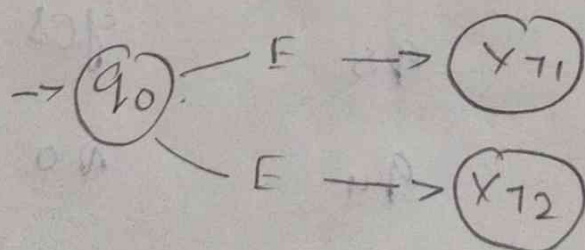
④ Regular language

Given L_1, L_2 are regular languages

$$M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$$

$$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$$

which accept L_1 & L_2



Final state

$$\text{are: } F = F_1 \cup F_2$$

$\therefore L_1 \cup L_2$ is regular

Hence, regular language are $F_1 = F_1 \cup F_2$

(11) $\{a^p \mid p \text{ is prime}\}$ regular?

The language is not regular.

Assume that C is regular

Choose a prime number p such that: $p > n$

Consider: $w = a^p$

Since p is prime: $w \in C$

By the pumping lemma $w = xyz$

where $|xy| \leq n$ & $|y| > 0$

Since the string only contains $y = a^k$

its length becomes $|x + p| = p + k$

$\therefore xy^{p+1}z \notin C$

Hence: $L = \{a^p \mid p \text{ is prime}\}$ is not regular

⑤ DFHA minimization

Initial = q_0 , Final = q_1, q_2, q_4

Non-final states = q_0, q_3, q_5

Transition table

State	0	1	Final
q_0	q_3	q_1	NO
q_1	q_2	q_5	YES
q_2	q_2	q_5	YES
q_3	q_0	q_4	NO
q_4	q_2	q_5	YES
q_5	q_5	q_5	NO

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Transition Table

minimized

$A = \{q_0, q_3\}$

$B = \{q_1, q_2, q_4\}$

$C = \{q_5\}$

0

A

B

C

1

B

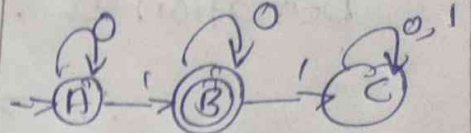
C

C

Initial state = A

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DF A (minimized)



$\{ \{q_0, q_3\}, \{q_1, q_2, q_4\}, \{q_5\} \}$